

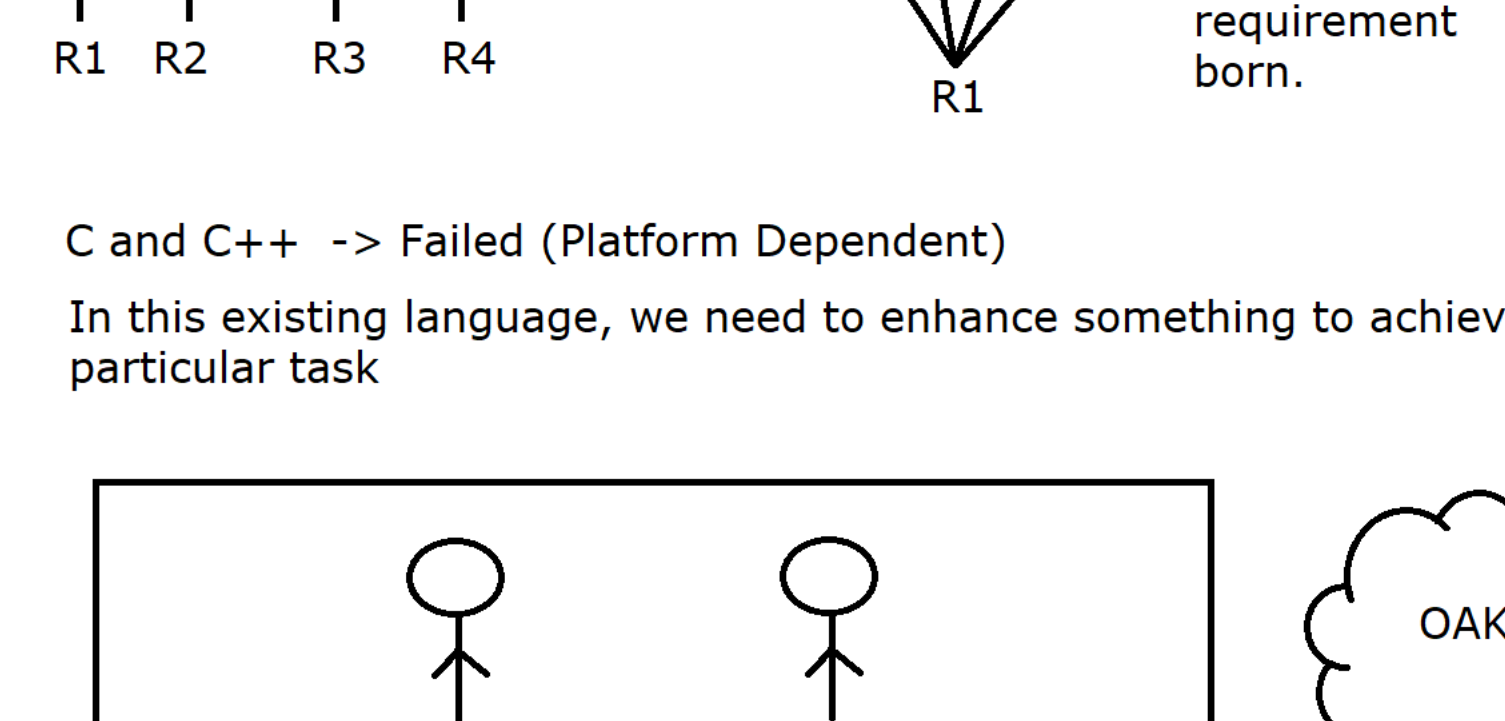
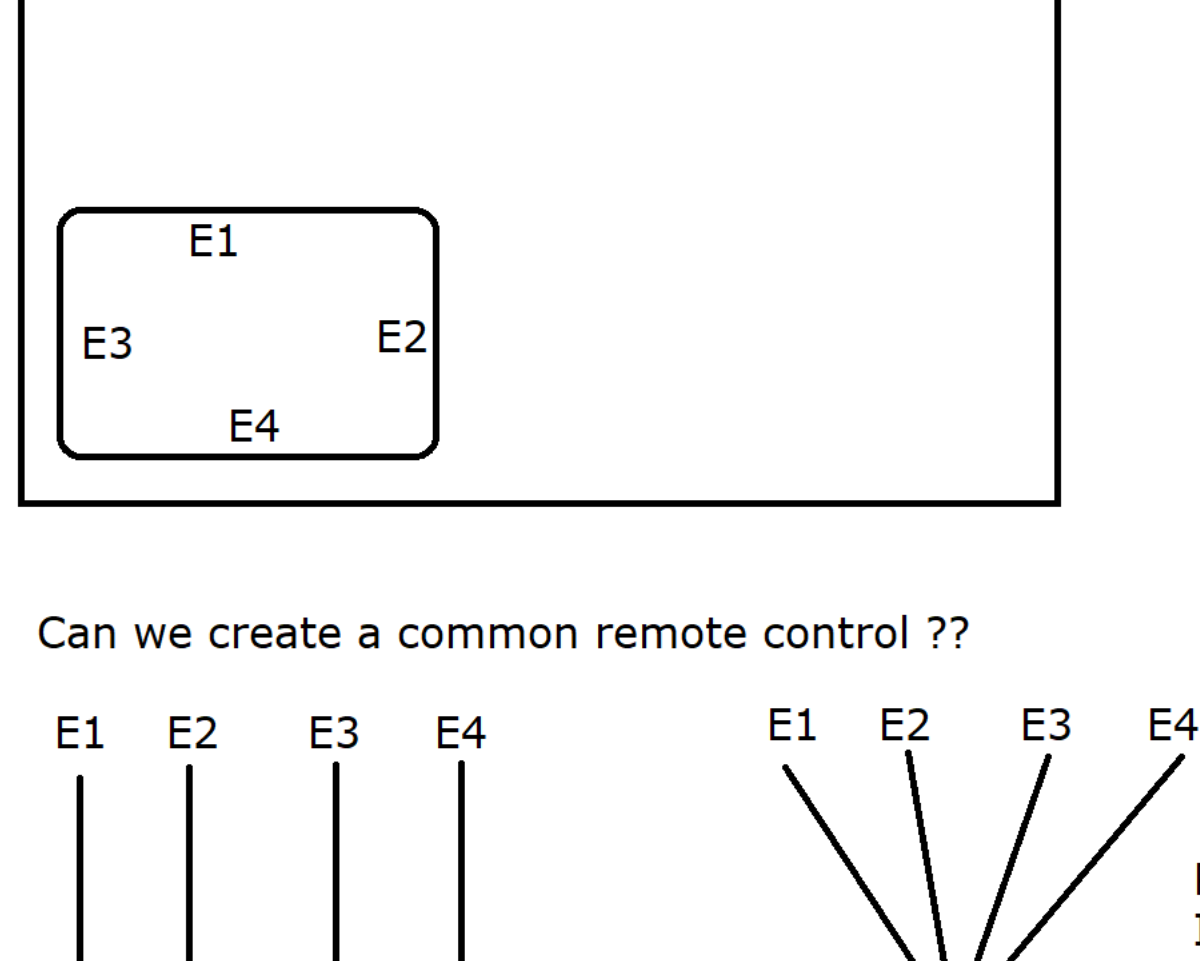
Applications of Java:

- a) Desktop OR Standalone Applications (e.g., IDEs, tools)
b) Web Applications
c) Mobile Applications (Android)
d) Enterprise Applications (banking, ERP, etc.)
e) Gaming Applications
f) Cloud-based and AI/ML Applications
g) Distributed Application

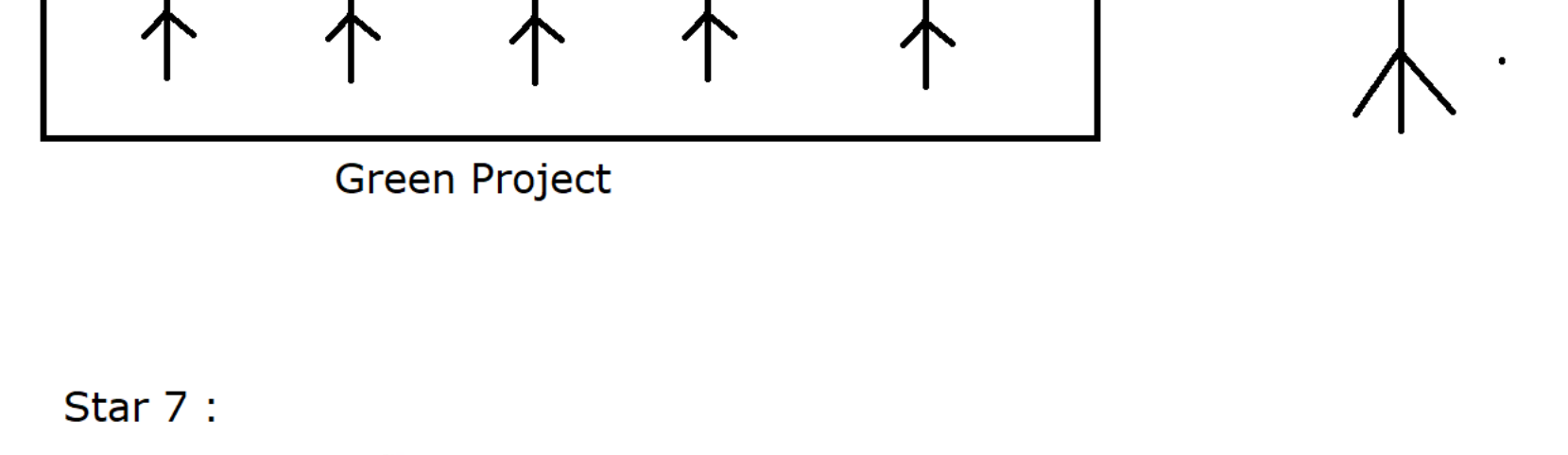
Editions of Java:

- a) Java SE (Java Standard Edition) Standalone/Desktop Apps
b) Java EE (Java Enterprise Edition) Web & Enterprise Apps
c) Java ME (Java Micro Edition) Mobile & Embedded Devices
d) JavaFX (Java FX) Rich Internet Apps (UI/Graphics) [Outdated]

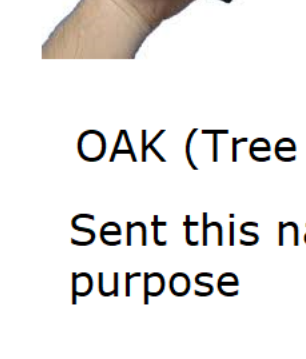
History of Java



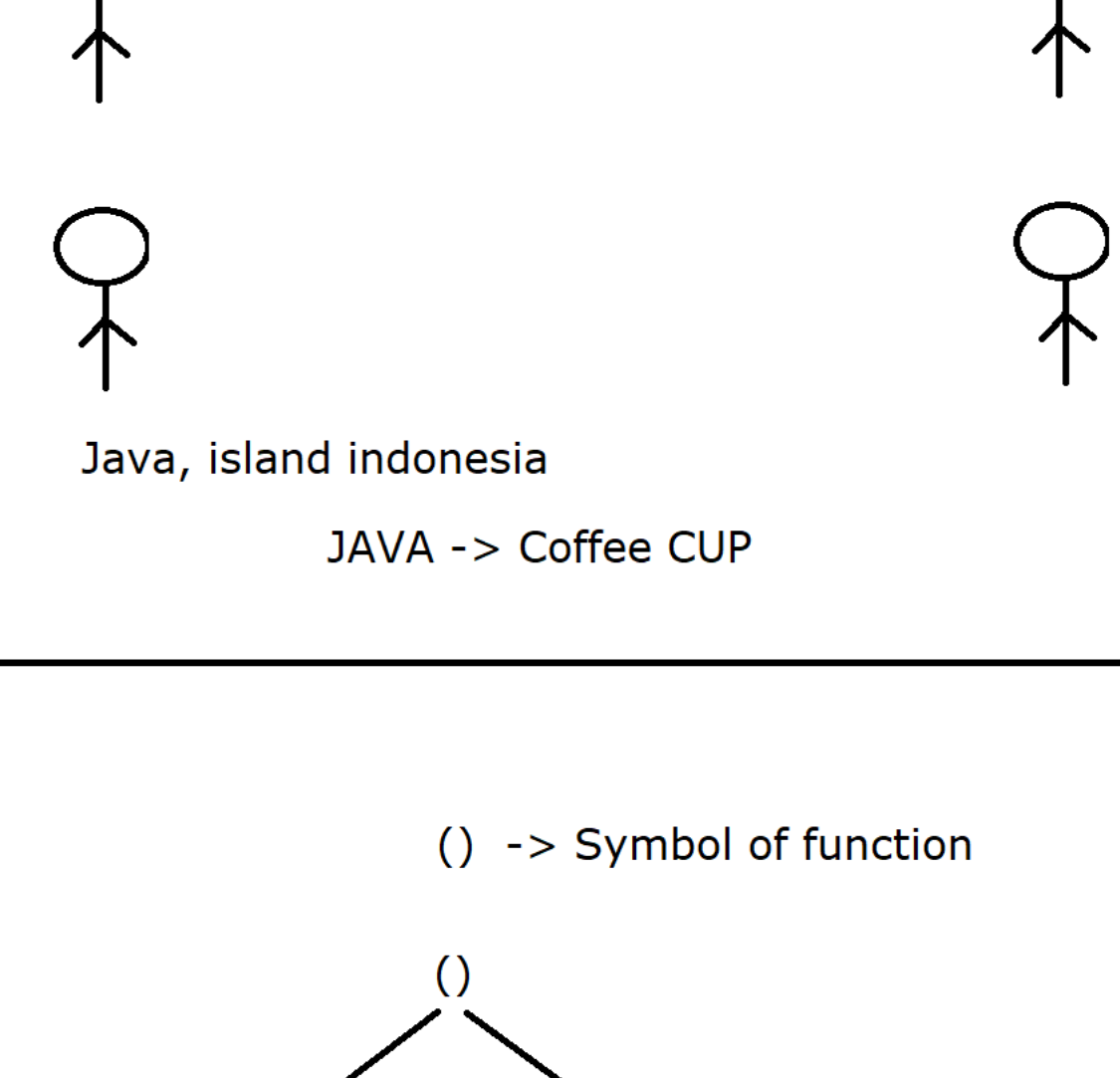
C and C++ -> Failed (Platform Dependent)
In this existing language, we need to enhance something to achieve this particular task



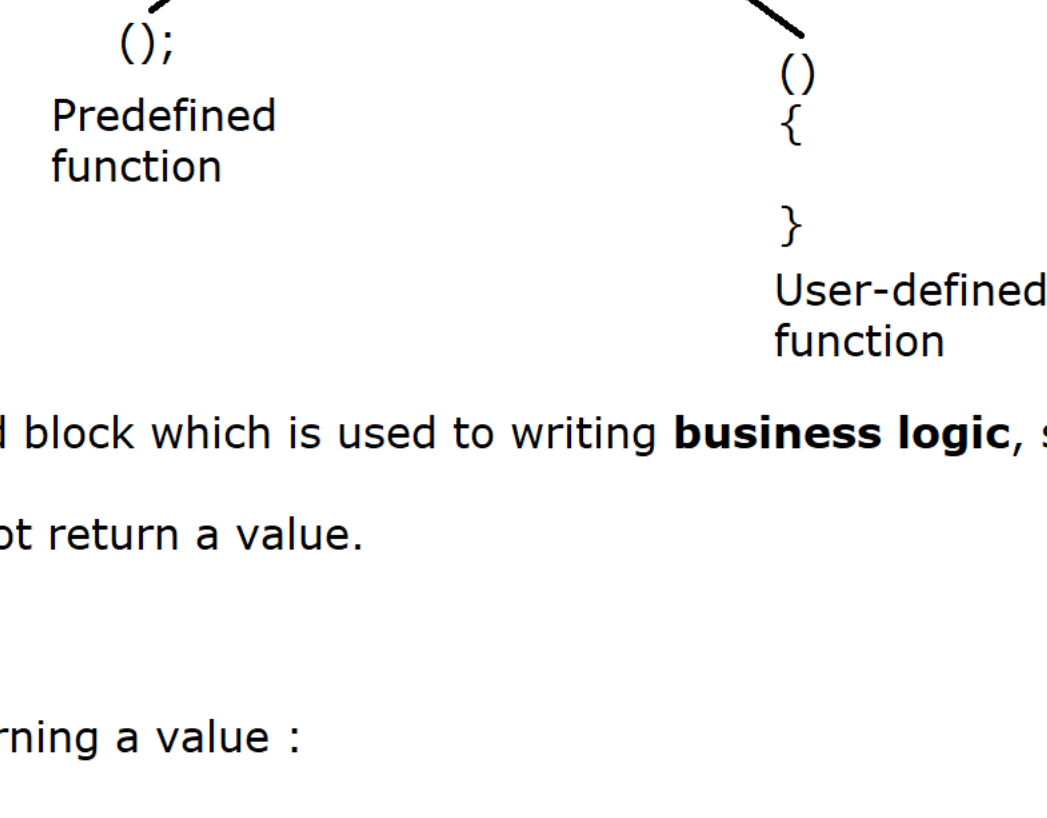
Star 7 :



OAK (Tree Name)
Sent this name to Lab for Registration purpose
Rename of OAK
=====



What is a function?



* A function is a self defined block which is used to writing **business logic**, some calculation OR printing the result.
* A function may OR may not return a value.

Case 1 :

A function which is not returning a value :

```
public void accept()  
{  
  
}
```

Case 2 :

A function which is returning an int value :

```
public int accept()  
{  
  
  return 9;  
}
```

A function is divided into two types:

- 1) Predefined OR Built-in function
2) User-defined OR Custom function

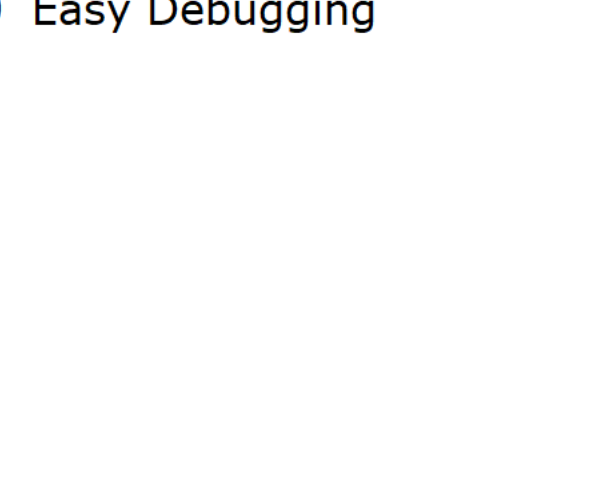
Predefined OR Built-in function

A function which is developed by language creator itself is called predefined function.

User-defined OR Custom Function

A function which is written by user/developer for its own requirement and specification is called User-defined function.

Advantages of a function :



Approach 1:

```
void main()  
{  
  int sum = a + b;  
  
  int sub = a - b;  
  
  int mul = a * b;  
  
  int div = a/b;  
  
}
```

- 1) Modularity
2) Easy to understand
3) Reusability
4) Easy Debugging

Approach 2:

```
void main()  
{  
  int sum(int x , int y)  
  {  
    return x + y;  
  }  
  
  int sub(int x , int y)  
  {  
    return x - y;  
  }  
  
  int mul(int x , int y)  
  {  
    return x * y;  
  }  
  
  int div(int x , int y)  
  {  
    return x / y;  
  }  
}
```